Experiment 8 - Autoencoders and Variational Autoencoders

Dataset: Fashion-MNIST
Framework: PyTorch

```
!pip install wandb huggingface_hub -q
```

```
import wandb
wandb.login()
```

```
/usr/local/lib/python3.12/dist-packages/notebook/notebookapp.py:191: SyntaxWarning: invalid escape sequence '\/'
  | |_| | '_ \/ _` / _` |  _/ -_)
wandb: (1) Create a W&B account
wandb: (2) Use an existing W&B account
wandb: (3) Don't visualize my results
wandb: Enter your choice: 2
wandb: You chose 'Use an existing W&B account'
wandb: Logging into https://api.wandb.ai. (Learn how to deploy a W&B server locally: https://wandb.me/wandb-server)
wandb: Create a new API key at: https://wandb.ai/authorize?ref=models
wandb: Store your API key securely and do not share it.
wandb: Paste your API key and hit enter: ··· ··· ··· ···
wandb: No netrc file found, creating one.
wandb: Appending key for api.wandb.ai to your netrc file: /root/.netrc
wandb: Currently logged in as: sundaramsahu111 (sundaramsahu111-dtu) to https://api.wandb.ai. Use `wandb login --relogin` to
True
```

```
from huggingface_hub import notebook_login
notebook_login()
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
```

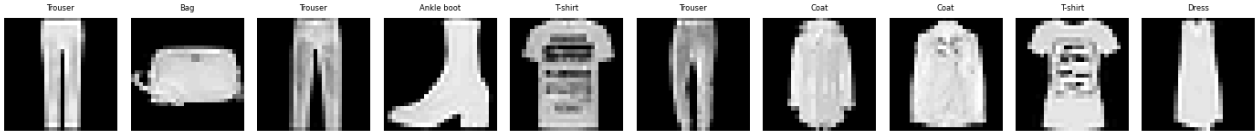
```
import torch
import torch.nn as nn
import torch.optim as optim
import torchvision
import torchvision.transforms as transforms
import matplotlib.pyplot as plt
import numpy as np
import wandb
from torch.utils.data import DataLoader, random_split
from huggingface_hub import HfApi, create_repo

# change this to your huggingface username
HF_REPO = "your-hf-username/exp8-ae-vae"

WANDB_PROJECT = "exp8-ae-vae"
EPOCHS       = 5
BATCH_SIZE   = 128
LATENT_DIMS  = [2, 8, 16, 32]
OPTIMIZERS   = ["adam", "sgd", "rmsprop"]
LOSS_FNS     = ["bce", "mse"]
DEVICE       = torch.device("cuda" if torch.cuda.is_available() else "cpu")

print(DEVICE)
```

```
cuda
```

```
transform = transforms.Compose([
    transforms.ToTensor()
])

full_train = torchvision.datasets.FashionMNIST(root='./data', train=True,  download=True, transform=transform)
test_data  = torchvision.datasets.FashionMNIST(root='./data', train=False, download=True, transform=transform)

n_train = int(0.8 * len(full_train))
n_val   = len(full_train) - n_train

train_set, val_set = random_split(full_train, [n_train, n_val], generator=torch.Generator().manual_seed(42))

train_loader = DataLoader(train_set, batch_size=BATCH_SIZE, shuffle=True,  num_workers=2)
val_loader   = DataLoader(val_set,   batch_size=BATCH_SIZE, shuffle=False, num_workers=2)
test_loader  = DataLoader(test_data, batch_size=BATCH_SIZE, shuffle=False, num_workers=2)

print(f'train: {len(train_set)}  val: {len(val_set)}  test: {len(test_data)}')

class_names = ['T-shirt','Trouser','Pullover','Dress','Coat','Sandal','Shirt','Sneaker','Bag','Ankle boot']

imgs, labels = next(iter(train_loader))
```

```
fig, axes = plt.subplots(1, 10, figsize=(14, 2))
for i in range(10):
    axes[i].imshow(imgs[i].squeeze(), cmap='gray')
    axes[i].set_title(class_names[labels[i]], fontsize=6)
    axes[i].axis('off')
plt.tight_layout()
plt.show()
```

```
100%|██████████| 26.4M/26.4M [00:00<00:00, 113MB/s]
100%|██████████| 29.5k/29.5k [00:00<00:00, 26.4MB/s]
100%|██████████| 4.42M/4.42M [00:00<00:00, 58.9MB/s]
100%|██████████| 5.15k/5.15k [00:00<00:00, 22.3MB/s]
train: 48000  val: 12000  test: 10000
```



Part A - Autoencoder

```
class Autoencoder(nn.Module):
    def __init__(self, latent_dim):
        super(Autoencoder, self).__init__()

        self.encoder = nn.Sequential(
            nn.Flatten(),
            nn.Linear(784, 512),
            nn.ReLU(),
            nn.Linear(512, 256),
            nn.ReLU(),
            nn.Linear(256, latent_dim)
        )

        self.decoder = nn.Sequential(
            nn.Linear(latent_dim, 256),
            nn.ReLU(),
            nn.Linear(256, 512),
            nn.ReLU(),
            nn.Linear(512, 784),
            nn.Sigmoid()
        )

    def forward(self, x):
        z = self.encoder(x)
        out = self.decoder(z).view(-1, 1, 28, 28)
        return out, z
```

Part B - Variational Autoencoder

```
class VAE(nn.Module):
    def __init__(self, latent_dim):
        super(VAE, self).__init__()
        self.latent_dim = latent_dim

        self.encoder = nn.Sequential(
            nn.Flatten(),
            nn.Linear(784, 512),
            nn.ReLU(),
            nn.Linear(512, 256),
            nn.ReLU()
        )

        self.fc_mu     = nn.Linear(256, latent_dim)
        self.fc_logvar = nn.Linear(256, latent_dim)

        self.decoder = nn.Sequential(
            nn.Linear(latent_dim, 256),
            nn.ReLU(),
            nn.Linear(256, 512),
            nn.ReLU(),
            nn.Linear(512, 784),
            nn.Sigmoid()
        )

    def encode(self, x):
        h = self.encoder(x)
        return self.fc_mu(h), self.fc_logvar(h)

    def reparameterize(self, mu, logvar):
        std = torch.exp(0.5 * logvar)
        eps = torch.randn_like(std)
        return mu + std * eps

    def decode(self, z):
        return self.decoder(z).view(-1, 1, 28, 28)

    def forward(self, x):
```

```
        mu, logvar = self.encode(x)
        z = self.reparameterize(mu, logvar)
        return self.decode(z), mu, logvar, z
```

```
def recon_loss(recon, x, loss_type):
    if loss_type == 'bce':
        return nn.BCELoss()(recon, x)
    return nn.MSELoss()(recon, x)

def vae_loss(recon, x, mu, logvar, loss_type):
    r_loss = recon_loss(recon, x, loss_type)
    kl     = -0.5 * torch.mean(1 + logvar - mu.pow(2) - logvar.exp())
    return r_loss + kl, r_loss, kl

def get_optimizer(model, name):
    if name == 'adam':
        return optim.Adam(model.parameters(), lr=1e-3)
    elif name == 'sgd':
        return optim.SGD(model.parameters(), lr=1e-3, momentum=0.9)
    else:
        return optim.RMSprop(model.parameters(), lr=1e-3)
```

```
def train_ae(model, loader, optimizer, loss_type):
    model.train()
    running = 0
    for x, _ in loader:
        x = x.to(DEVICE)
        optimizer.zero_grad()
        out, _ = model(x)
        loss = recon_loss(out, x, loss_type)
        loss.backward()
        optimizer.step()
        running += loss.item()
    return running / len(loader)

def evaluate_ae(model, loader, loss_type):
    model.eval()
    running = 0
    with torch.no_grad():
        for x, _ in loader:
            x = x.to(DEVICE)
            out, _ = model(x)
            running += recon_loss(out, x, loss_type).item()
    return running / len(loader)

def train_vae(model, loader, optimizer, loss_type):
    model.train()
    t_loss = t_recon = t_kl = 0
    for x, _ in loader:
        x = x.to(DEVICE)
        optimizer.zero_grad()
        out, mu, logvar, _ = model(x)
        loss, rl, kl = vae_loss(out, x, mu, logvar, loss_type)
        loss.backward()
        optimizer.step()
        t_loss += loss.item()
        t_recon += rl.item()
        t_kl += kl.item()
    n = len(loader)
    return t_loss/n, t_recon/n, t_kl/n

def evaluate_vae(model, loader, loss_type):
    model.eval()
    t_loss = t_recon = t_kl = 0
    with torch.no_grad():
        for x, _ in loader:
            x = x.to(DEVICE)
            out, mu, logvar, _ = model(x)
            loss, rl, kl = vae_loss(out, x, mu, logvar, loss_type)
            t_loss += loss.item()
            t_recon += rl.item()
            t_kl += kl.item()
    n = len(loader)
    return t_loss/n, t_recon/n, t_kl/n
```

```
def show_reconstructions(model, loader, model_type, title, n=8):
    model.eval()
    x, _ = next(iter(loader))
    x = x[:n].to(DEVICE)
    with torch.no_grad():
        out = model(x)[0]
    fig, axes = plt.subplots(2, n, figsize=(n * 1.4, 3))
    for i in range(n):
        axes[0, i].imshow(x[i].cpu().squeeze(), cmap='gray')
        axes[1, i].imshow(out[i].cpu().squeeze(), cmap='gray')
        axes[0, i].axis('off')
        axes[1, i].axis('off')
    axes[0, 0].set_ylabel('Input', fontsize=8)
    axes[1, 0].set_ylabel('Recon', fontsize=8)
    plt.suptitle(title, fontsize=9)
    plt.tight_layout()
    plt.show()
    return fig

def show_latent_2d(model, loader, model_type, title):
```

```
    model.eval()
    zs, ys = [], []
    with torch.no_grad():
        for x, y in loader:
            x = x.to(DEVICE)
            z = model(x)[1] if model_type == 'AE' else model.encode(x)[0]
            zs.append(z.cpu().numpy())
            ys.append(y.numpy())
    zs = np.concatenate(zs)
    ys = np.concatenate(ys)
    fig, ax = plt.subplots(figsize=(7, 5))
    sc = ax.scatter(zs[:, 0], zs[:, 1], c=ys, cmap='tab10', s=2, alpha=0.5)
    cb = plt.colorbar(sc, ax=ax)
    cb.set_ticks(range(10))
    cb.set_ticklabels(class_names)
    ax.set_title(title)
    ax.set_xlabel('z0')
    ax.set_ylabel('z1')
    plt.tight_layout()
    plt.show()
    return fig

def show_interpolation(model, loader, model_type, title, steps=10):
    model.eval()
    x, _ = next(iter(loader))
    x1 = x[0:1].to(DEVICE)
    x2 = x[1:2].to(DEVICE)
    with torch.no_grad():
        z1 = model(x1)[1] if model_type == 'AE' else model.encode(x1)[0]
        z2 = model(x2)[1] if model_type == 'AE' else model.encode(x2)[0]
    fig, axes = plt.subplots(1, steps, figsize=(steps * 1.3, 1.8))
    for i, a in enumerate(np.linspace(0, 1, steps)):
        z = (1 - a) * z1 + a * z2
        with torch.no_grad():
            img = model.decoder(z).view(28, 28) if model_type == 'AE' else model.decode(z).squeeze()
        axes[i].imshow(img.cpu(), cmap='gray')
        axes[i].set_title(f'{a:.1f}', fontsize=6)
        axes[i].axis('off')
    plt.suptitle(title, fontsize=9)
    plt.tight_layout()
    plt.show()
    return fig

def show_generated(model, title, n=10):
    model.eval()
    with torch.no_grad():
        z = torch.randn(n, model.latent_dim).to(DEVICE)
        imgs = model.decode(z)
    fig, axes = plt.subplots(1, n, figsize=(n * 1.3, 1.8))
    for i in range(n):
        axes[i].imshow(imgs[i].cpu().squeeze(), cmap='gray')
        axes[i].axis('off')
    plt.suptitle(title, fontsize=9)
    plt.tight_layout()
    plt.show()
    return fig
```

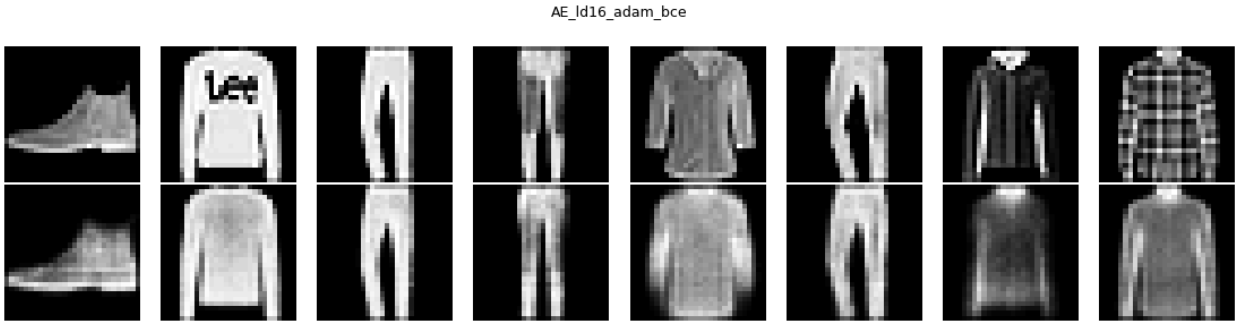
```
results = {}
trained_models = {}

def run_experiment(model_type, latent_dim, opt_name, loss_type):
    name = f'{model_type}_ld{latent_dim}_{opt_name}_{loss_type}'
    print(f'\nstarting: {name}')

    wandb.init(project=WANDB_PROJECT, name=name, config={
        'model': model_type,
        'latent_dim': latent_dim,
        'optimizer': opt_name,
        'loss': loss_type,
        'epochs': EPOCHS
    })

    model     = (Autoencoder(latent_dim) if model_type == 'AE' else VAE(latent_dim)).to(DEVICE)
    optimizer = get_optimizer(model, opt_name)

    for epoch in range(1, EPOCHS + 1):
        if model_type == 'AE':
            tr = train_ae(model, train_loader, optimizer, loss_type)
            vl = evaluate_ae(model, val_loader, loss_type)
            wandb.log({'epoch': epoch, 'train_loss': tr, 'val_loss': vl})
            print(f'  epoch {epoch}/{EPOCHS}  train={tr:.4f}  val={vl:.4f}')
        else:
            tr, tr_r, tr_k = train_vae(model, train_loader, optimizer, loss_type)
            vl, vl_r, vl_k = evaluate_vae(model, val_loader, loss_type)
            wandb.log({
                'epoch': epoch,
                'train_loss': tr, 'train_recon': tr_r, 'train_kl': tr_k,
                'val_loss':   vl, 'val_recon':   vl_r, 'val_kl':   vl_k
            })
            print(f'  epoch {epoch}/{EPOCHS}  train={tr:.4f}  recon={tr_r:.4f}  kl={tr_k:.4f}  val={vl:.4f}')

    if model_type == 'AE':
        test_loss = evaluate_ae(model, test_loader, loss_type)
    else:
        test_loss, _, _ = evaluate_vae(model, test_loader, loss_type)

    print(f'  test_loss={test_loss:.4f}')
```

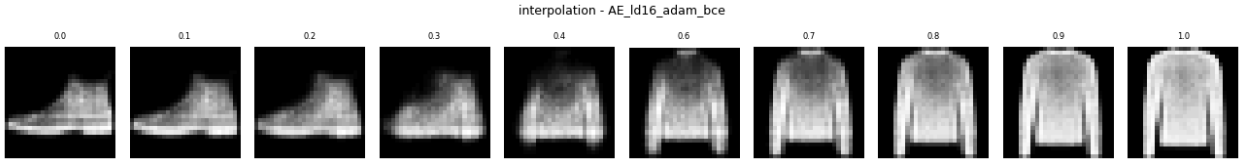
```
    wandb.log({'test_loss': test_loss})

    fig = show_reconstructions(model, test_loader, model_type, name)
    wandb.log({'reconstructions': wandb.Image(fig)})
    plt.close(fig)

    fig = show_interpolation(model, test_loader, model_type, f'interpolation - {name}')
    wandb.log({'interpolation': wandb.Image(fig)})
    plt.close(fig)

    if latent_dim == 2:
        fig = show_latent_2d(model, test_loader, model_type, f'latent 2D - {name}')
        wandb.log({'latent_2d': wandb.Image(fig)})
        plt.close(fig)

    if model_type == 'VAE':
        fig = show_generated(model, f'generated - {name}')
        wandb.log({'generated': wandb.Image(fig)})
        plt.close(fig)

    torch.save(model.state_dict(), f'{name}.pt')
    trained_models[name] = model
    results[name] = test_loss

    wandb.finish()
    return model
```

Quick Test - run this first

```
run_experiment('AE',  latent_dim=16, opt_name='adam', loss_type='bce')
run_experiment('VAE', latent_dim=16, opt_name='adam', loss_type='bce')
```

```
starting: AE_ld16_adam_bce
Tracking run with wandb version 0.25.1
Run data is saved locally in /content/wandb/run-20260416_022525-kgd8c2qt
Syncing run AE_ld16_adam_bce to Weights & Biases (docs)
View project at https://wandb.ai/sundaramsahu111-dtu/exp8-ae-vae
View run at https://wandb.ai/sundaramsahu111-dtu/exp8-ae-vae/runs/kgd8c2qt
  epoch 1/5  train=0.3440  val=0.3051
  epoch 2/5  train=0.2979  val=0.2938
  epoch 3/5  train=0.2896  val=0.2874
  epoch 4/5  train=0.2851  val=0.2837
  epoch 5/5  train=0.2821  val=0.2820
  test_loss=0.2836
```





Run history:

epoch	▁▅▆▇▇
test_loss	▁
train_loss	▇▅▃▂▁
val_loss	▇▅▃▂▁

Run summary:

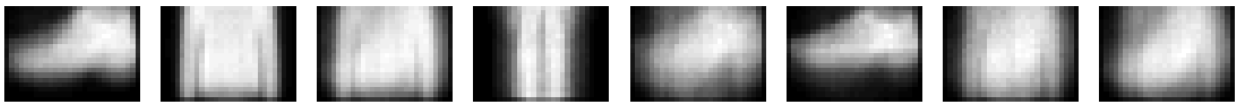
epoch	5
test_loss	0.28359
train_loss	0.28213
val_loss	0.28203

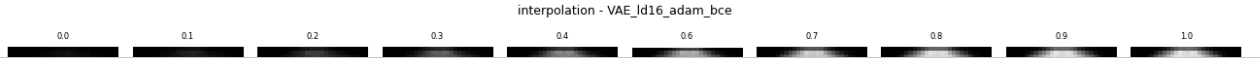
View run AE_ld16_adam_bce at: https://wandb.ai/sundaramsahu111-dtu/exp8-ae-vae/runs/kgd8c2qt
View project at: https://wandb.ai/sundaramsahu111-dtu/exp8-ae-vae
Synced 5 W&B file(s), 2 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: `./wandb/run-20260416_022525-kgd8c2qt/logs`

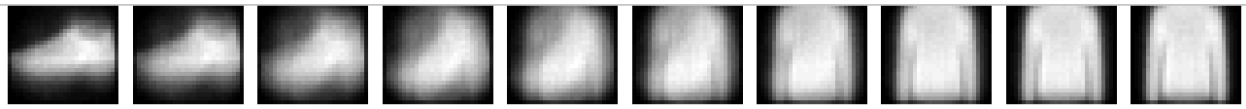
```
starting: VAE_ld16_adam_bce
Tracking run with wandb version 0.25.1
Run data is saved locally in /content/wandb/run-20260416_022614-x5zgivrw
Syncing run VAE_ld16_adam_bce to Weights & Biases (docs)
View project at https://wandb.ai/sundaramsahu111-dtu/exp8-ae-vae
View run at https://wandb.ai/sundaramsahu111-dtu/exp8-ae-vae/runs/x5zgivrw
  epoch 1/5  train=0.4942  recon=0.4880  kl=0.0062  val=0.4883
  epoch 2/5  train=0.4839  recon=0.4615  kl=0.0224  val=0.4813
  epoch 3/5  train=0.4791  recon=0.4432  kl=0.0360  val=0.4797
  epoch 4/5  train=0.4778  recon=0.4354  kl=0.0425  val=0.4780
  epoch 5/5  train=0.4771  recon=0.4302  kl=0.0469  val=0.4779
  test_loss=0.4777
```

Full Grid - all 48 combinations

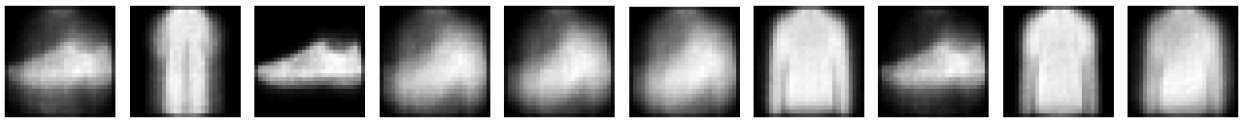
```
for model_type in ['AE', 'VAE']:
    for ld in LATENT_DIMS:
        for opt in OPTIMIZERS:
            for lf in LOSS_FNS:
                run_experiment(model_type, ld, opt, lf)
```







generated - VAE_ld16_adam_bce



Run history:

epoch	▁▂▃▅▆▇▇▇▇▇
test_loss	▁
train_kl	▁▂▃▅▆▇▇▇▇▇
train_loss	▇▆▅▃▂▁▁▁▁▁
train_recon	▇▆▅▃▂▁▁▁▁▁
val_kl	▁▂▃▅▆▇▇▇▇▇
val_loss	▇▆▅▃▂▁▁▁▁▁
val_recon	▇▆▅▃▂▁▁▁▁▁

Run summary:

epoch	5
test_loss	0.47766
train_kl	0.04691
train_loss	0.47709
train_recon	0.43019
val_kl	0.04706
val_loss	0.47795
val_recon	0.43089

View run **VAE_ld16_adam_bce** at: https://wandb.ai/sundaramsahu111-dtu/exp8-ae-vae/runs/x5zgivrw
View project at: https://wandb.ai/sundaramsahu111-dtu/exp8-ae-vae
Synced 5 W&B file(s), 3 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: `./wandb/run-20260416_022614-x5zgivrw/logs`

```
VAE(
  (encoder): Sequential(
    (0): Flatten(start_dim=1, end_dim=-1)
    (1): Linear(in_features=784, out_features=512, bias=True)
    (2): ReLU()
    (3): Linear(in_features=512, out_features=256, bias=True)
    (4): ReLU()
  )
  (fc_mu): Linear(in_features=256, out_features=16, bias=True)
  (fc_logvar): Linear(in_features=256, out_features=16, bias=True)
  (decoder): Sequential(
    (0): Linear(in_features=16, out_features=256, bias=True)
    (1): ReLU()
    (2): Linear(in_features=256, out_features=512, bias=True)
    (3): ReLU()
    (4): Linear(in_features=512, out_features=784, bias=True)
    (5): Sigmoid()
  )
)
```